

Your submitted solution is only **valid** if it contains the following paragraphs at the top of the program. Please *fill* in *your name* and *Neptun code*.

```
/*<NAME>, <NEPTUN>
```

```
This solution was prepared and submitted by the student stated above for the assignment of the Programming course. I declare that this solution is my own work. I have not copied or used third party solutions. I have not passed my solution to my classmates, neither made it public.
```

```
Students' regulation of Eötvös Loránd University (ELTE Regulations Vol. II. 74/C.§) states that as long as a student presents another student's work - or at least the significant part of it - as his/her own performance, it will count as a disciplinary fault. The most serious consequence of a disciplinary fault can be dismissal of the student from the University.*//
```

When submitting on TMS upload the program with the output enabled that only prints the result of the specific subtask e.g.: if you are solving a) do not include the outputs for tasks b), c), d)!

Hogwarts House Challenges

The Hogwarts professors are evaluating each house's performance in a magical obstacle course. Each row in the matrix represents a challenge, and the columns represent the performance (in points) of the houses in that challenge. The professors are analyzing the results to ensure fair competition and identify standout performances.



Input

The first line contains two integers N ($1 \leq N \leq 50$) and M ($1 \leq M \leq 50$), representing the number of rows (N challenges) and columns (M houses) in the matrix.

The next N lines contain M integers each, representing the performance of houses in the respective challenge ($1 \leq \text{value} \leq 100$).

Write a program that solves the following tasks:

Tasks:

- Print the smallest performance value across the entire matrix.
- Calculate and print the total performance points across all challenges and houses.
- For each challenge (row), find the house with the highest performance and print the maximum value for each row in the order they appear.
- Count and print how many houses have an average performance that is equal or greater than 80%. After that, list each qualifying house with its index (1-based).

Output

Each task's output should be separated by new lines. Do *NOT* print the text in red; the red color is used here solely to illustrate and distinguish each subtask's output for clarity.

Example

Input

```
4 3
60 85 90
95 80 75
100 85 90
90 70 80
```

Output

```
Task a):
60

Task b):
1000

Task c):
90 95 100 90

Task d):
2
2 3
```